Assignment 11 – Ethics and Responsible-Use Statement

Team Members: Nate Heath, Elliot Warner, William Thomas

Project Title: Data-Day

Goal Statement: To build and open source a 64 x 32 pixel desktop display and the developer platform for creating applications for the device.

Privacy Summary

Data item	Source	Interface	Purpose	Consent	Storage	Access	Retention
User Python Script	Local machine	I5 (Local Machine -> Upload Server)	Execution on Orange Pi to render visuals/apps on 64x32 LED matrix (REQ-01, REQ-05)	Explicit Opt-in via Web UI	Stored unencrypted on Orange Pi microSD card	Accessible only by local system user and device administrator	Retained until explicitly deleted
Upload Payload Metadata	Local machine request headers	I5 (Local Machine -> Upload Server)	API response formatting, and slot assignment (REQ-06)	Explicit Opt-in	Stored in local JSON registry file	Read/Write access restricted strictly to the FastAPI backend	Deleted when the corresponding application file is removed
Sanitization & Security Inspection Status	Security scanner	I6 (Upload Server -> Device Application)	Validating code safety (NFR-02) and blocking unsafe imports	Automated System Pre-requisite	In-memory during AST evaluation. Pass/fail boolean logged to system	Readable by device log watcher and administrator	Log entries rotated automatically, lasts 30 days stored locally
Dial Rotation State	Rotary encoder hardware	I1 (Dial -> Orange Pi GPIO)	Navigation of application selection view (REQ-02) and user app inputs	Implicit via Physical Action	Transient in-memory state inside Linux kernel GPIO buffer	Hardware kernel driver	Discarded immediately after state transition event handling
Execution logs	Orange Pi execution manager	I2 (Orange Pi -> Device Application)	Debugging, system recovery and crash tracking	Operational Necessity	Plaintext file written to Orange Pi microSD card	System user	Stored locally for 30 days
Framebuffer	SDK / Application runtime	I4 (Device Application -> Display Matrix)	Driving physical HUB75 LED display panel hardware via	Implicit via Operation	Orange Pi volatile memory	Driver thread	Wiped immediately if exiting app or the system is shutdown

			memory mapping				

Explicitly Removed

Supabase: Originally, we planned on using a Supabase database and cloud telemetry pipeline for online storage and tracking. We have since removed it since the product is a single-owner local hardware device and the applications can be stored/installed on GitHub.

Security Summary

Default Credentials:

Risk: Computers like the Orange Pi Zero 3 often ship with default operating system credentials.

Mitigation: To access the repository and install the software on the Orange Pi, the user must already be signed in.

Unauthenticated Endpoints:

Risk: The local REST API allows code uploads, which could permit malicious lateral network attacks.

Mitigation: API key authentication and local subnet pinning. Users will have a secret local token that must be authenticated when using the API. These tokens will only be valid for the lifetime of the session. Also, the FastAPI server binds exclusively to the local network interface and will reject external requests.

Secrets in Source Control:

Risk: Committing SSH keys, WiFi passwords or API tokens to the repository.

Mitigation: Implement strict .gitignore rules to prevent environment files from entering version control.

Repository Git History Audit Results

```
\Capstone_2027> git log -p -S "SECRET" -S "PASSWORD" -S "PRIVATE KEY" -S "api_key"
\Capstone_2027>
```

Self-Audit Questions and Answers

Self-Audit Question	Team answer
1) Are default passwords and keys removed?	Yes
2) Are all endpoints authenticated or locally bound?	Yes
3) Has the full git history been checked for exposed secrets?	Yes
4) Is user-submitted code isolated from host OS system files?	Yes
5) Are we preventing uploaded scripts from filling memory/storage?	Yes

Licensing Summary

No dependencies are installed yet. Versions named in the Test Plan are listed; everything else is unpinned and will be pinned in requirements.txt / package.json at the first code commit, with this table updated.

Name	Version	License (SPDX)	Type	Obligation
FastAPI (upload server)	unpinned	MIT	Permissive	Keep copyright and license notice
Uvicorn (ASGI server)	unpinned	BSD-3-Clause	Permissive	Keep notice; no endorsement using their name
python-multipart (FastAPI file uploads)	unpinned	Apache-2.0	Permissive	Keep notice and NOTICE file; state changes if modified
React (upload page)	unpinned	MIT	Permissive	Keep notice

Node.js (build only, not shipped)	20 LTS	MIT	Permissive	None for our release
Python	3.11	PSF-2.0	Permissive	Keep notice if redistributed in an image
pytest (test only)	8	MIT	Permissive	None (not shipped)
pytest-cov (test only)	unpinned	MIT	Permissive	None (not shipped)
httpx (test only)	unpinned	BSD-3-Clause	Permissive	None (not shipped)
HUB75 display driver (not yet chosen; any compatible HUB75 LED panel supported)	n/a	Must be MIT, BSD, or Apache-2.0; LGPL only if dynamically linked and unmodified	Permissive required	Keep notice; a GPL driver may not be used in the same process as our code
GPIO library for dial (not yet chosen)	n/a	Must be MIT, BSD, or Apache-2.0; LGPL only if dynamically linked and unmodified	Permissive required	Keep notice
Armbian OS / Linux kernel	Debian 12 Bookworm	GPL-2.0-only (kernel) and others	Copyleft	Separate program, not linked with our code (mere aggregation); if we publish a full image, offer corresponding source
Datasets	none	n/a	n/a	None used; demo apps calling weather / UC shuttle APIs will add those services' terms here

Fonts, icons, images	none yet	n/a	n/a	Bitmap font for text rendering (task 13) will be added with its license

Project license: MIT (LICENSE file at the repository root).

Compatibility: Every permissive row above (MIT, BSD-3-Clause, Apache-2.0, PSF-2.0) is compatible with an MIT project; we keep each one's copyright and license notice in NOTICE.md. Test-only and build-only tools are not distributed with the project. Armbian and the Linux kernel are copyleft, but they are a separate operating system our code runs on rather than code we link to or modify, so they place no license obligation on Data-Day's source.

Copyleft flag and decision: The most widely used HUB75 driver, rpi-rgb-led-matrix, is GPL-2.0-or-later. Linking it into our SDK would require releasing Data-Day under the GPL, which conflicts with our MIT license. Because Data-Day is meant to work with any compatible HUB75 LED panel, we are not tied to that driver. We decided that the display driver and GPIO library must be permissively licensed (or LGPL, dynamically linked and unmodified). Any GPL driver is excluded

Attribution List

This same list is maintained in NOTICE.md at the repository root.

Libraries

Library	Version	Author / Maintainer	Source URL	License
FastAPI	unpinned	Sebastián Ramírez and contributors	https://github.com/fastapi/fastapi	MIT
Uvicorn	unpinned	Encode OSS and contributors	https://github.com/encode/uvicorn	BSD-3-Clause
python-multipart	unpinned	Andrew Dunham, Marcelo Trylesinski and contributors	https://github.com/Kludex/python-multipart	Apache-2.0

React	unpinned	Meta Platforms, Inc. and contributors	https://github.com/facebook/react	MIT
Node.js (build only)	20 LTS	OpenJS Foundation and contributors	https://github.com/nodejs/node	MIT
Python	3.11	Python Software Foundation	https://www.python.org	PSF-2.0
pytest (test only)	8	pytest-dev	https://github.com/pytest-dev/pytest	MIT
pytest-cov (test only)	unpinned	pytest-dev	https://github.com/pytest-dev/pytest-cov	MIT
httpx (test only)	unpinned	Encode OSS and contributors	https://github.com/encode/httpx	BSD-3-Clause
Armbian (operating system)	Debian 12 Bookworm	Armbian project	https://www.armbian.com	Various; Linux kernel GPL-2.0-only
HUB75 display driver	not yet chosen	TBD	TBD	Must be MIT, BSD, or Apache-2.0 (see Licensing Summary)
GPIO library for dial	not yet chosen	TBD	TBD	Must be MIT, BSD, or Apache-2.0 (see Licensing Summary)

Code adapted from tutorials or other repositories

None so far. Any snippet adapted from a tutorial, Stack Overflow, or another repository will carry a comment above it with the source URL and license and will be added to this list in the same pull request.

Datasets

None used. If demo applications call outside services (for example weather or UC shuttle data for US01), each service will be listed here with how its data is collected and its usage restrictions before the demo ships.

Non-code assets (icons, fonts, images)

None so far. The bitmap font needed for text rendering (task 13) will be added here with its author and license before it is committed.

AI-Use Disclosure

"Generative AI tools may be used for drafting and code assistance but not for the design decisions or written analysis that are assessed in this course."

Tool	Used by	Task	Column	How verified
Claude (Anthropic)	Elliot Warner	Looked up OS default credentials (Armbian, Orange Pi images) and the SPDX licenses of planned dependencies	Generally OK	Checked each claim against the project's own documentation or license file
ChatGPT	Nate Heath	General architecture planning and ensuring components cohesively work together	Generally OK	Checked responses against technical component specs and documentation
Copilot	William Thomas	Writing repetitive unit test cases for SDK pixel operations	Generally OK	Code-reviewed line-by-line and verified by running them

Confirmation. Each member confirms that the architecture, data model, priorities, and design rationale in this project were the team's own.

- ☐ Nate Heath: "I confirm the architecture, data model, priorities, and design rationale were the team's own." Nate Heath, Date: 11/11/2026

- Elliot Warner: "I confirm the architecture, data model, priorities, and design rationale were the team's own." Elliot Warner, Date: 11/11/2026
- William Thomas: "I confirm the architecture, data model, priorities, and design rationale were the team's own." William Thomas, Date: 11/11/2026